

Empirical Research in Finance

6. Application: Leverage and Banker-CEO Networks

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What we're about to do

Research question. *Do banker–CEO networks affect firm leverage?*

The dataset (data1.dta)- artificial data. US firm-year panel, 2000–2014. Accounting and market variables + network (banker–CEO connection measure).

The main regression we will build, in four steps:

$$\text{lev}_{it} = \alpha_i + \delta_t + \beta \text{network}_{it} + \gamma' X_{it} + \varepsilon_{it}$$

with firm FE (α_i), year FE (δ_t), controls

$X = \{\ln_age, \ln_nseg, mb, fcf, roa\}$, and SE clustered by firm.

Two key terms — in plain English

Leverage = how much of a firm's assets are financed by debt rather than equity.

Network = how many bankers the CEO personally knows (a measure of the CEO's banking connections).

The dataset — what's inside data1.dta

7,426 firm-year observations, 19 variables, US firms, 2000–2014. Each row = one firm in one fiscal year, identified by (firmid, fyear).

Role	Variables
Identifiers	firmid · fyear (fiscal year)
Universe	fic (country of incorporation) · sic (industry code)
Accounting (raw \$)	at assets · capx capex · ceq equity · csho shares · dlc short-term debt · dltd long-term debt · ebitda · rdip in-process R&D
Market	prcc_f (fiscal-year-end close price)
Firm descrip- tors	age (years listed) · nseg (# business segments)
Main x	network (avg # of banker connections to the CEO)

The five steps

Every empirical paper follows the same five steps:

- 1. FILTER** define a usable sample
- 2. CREATE VARS** log skewed vars/ ratios
- 3. WINSORIZE** cap tails at 1%/99%
- 4. DESCRIBE** means, sd, correlations, univariate test
- 5. ESTIMATE** pooled OLS → +controls/FE → +cluster
→ firm FE

STEP 1 — Filter

Filter sparsely. Every choice must be defensible in writing.

- ▶ drop missing or non-positive total assets
- ▶ drop financials (SIC 6000–6999)
- ▶ restrict geography
- ▶ *additional filters: stay sparing.* Check descriptives FIRST, then decide — each extra cut costs N and invites cherry-picking criticism

STEPS 2–3 — Create variables and winsorize

Step 2 — Create variables on the filtered sample:

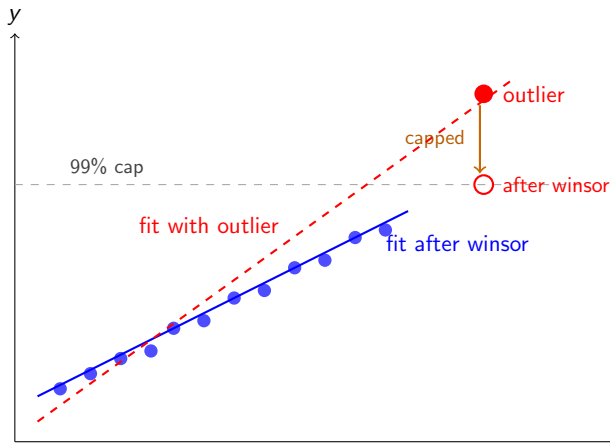
- ▶ many variables are defined from raw data
- ▶ sometimes it is useful to create logs, e.g, right-skewed counts (firm age, segments)

Step 3 — Winsorize (mostly) ratios:

- ▶ caps extremes without dropping observations — keeps your sample size.
- ▶ use the standard 1%/99% cut — choosing a non-standard threshold forces you to justify it
- ▶ because ratios with small denominators (e.g. ROA when assets are tiny) blow up to absurd values

The effect of outliers

One extreme observation can change the slope of an OLS regression.



A single extreme point (red) tilts the OLS line away from the bulk of the data. Capping the variable keeps the observation in the sample but pulls it back toward the body of the distribution — recovering the blue fit.

STEP 4 — Describe (before you estimate)

- ▶ Define ONE sample indicator; every regression in the table uses the same N
- ▶ **Summary statistics** — mean, sd, p_{10} , median, p_{90} , min, max, N for every variable that enters the regression.
- ▶ **Pairwise correlations** — spot multicollinearity, check signs before estimating.
- ▶ **Univariate test** — split on the dependent variable, t -test the main x . A first unconditional answer.

STEP 5 — The upgrade ladder

(1) Pooled OLS	$y_{it} = \alpha + \beta x_{it} + \varepsilon_{it}$
(2) +Controls + Year/Industry FE	add omitted controls; absorb common shocks
(3) +Clustered SE (firm)	residuals are correlated within firm
(4) Firm FE + Clustered SE	uses within-firm variation over time

Before we touch the data — make a prediction

Do you expect more-connected CEOs to borrow
more, **less**, or **about the same**?

- ▶ More? — bankers trust people they know; cheaper / easier credit.
- ▶ Less? — well-connected CEOs can raise equity more easily, or care more about reputation and pick safer capital structures.
- ▶ About the same? — the network may matter for *terms* (rate, covenants) but not for the *amount*.

Why predict first

Writing down your prior makes the regression result *informative*.
We will come back to this when we see the results.